

Single-cell transcriptomics reveals a cross-cohort decidual inflammatory macrophage/APC signature in recurrent pregnancy loss

Short title: Cross-cohort decidual inflammatory macrophage/APC signature in RPL

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Abstract

Recurrent pregnancy loss is a clinically heterogeneous reproductive disorder and a major reproductive-health problem. A substantial fraction of affected patients remain without a definitive explanation after evaluation for chromosomal, anatomical, endocrine, thrombotic and immune contributors, while public transcriptomic studies remain difficult to compare because decidual bulk profiles mix cell abundance, cell state and sampling differences. Here, we used a single-cell transcriptomic cross-cohort recurrence design to test whether recurrent-miscarriage-associated tissue-state signatures derived from a disease single-cell/pseudobulk reference recur across independent bulk decidua cohorts. GSE214607 was used to derive compact projection-ready signatures from first-trimester decidua and villi samples. Four signatures were projected into three independent bulk decidua datasets, GSE161969, GSE113790 and GSE178535, using within-cohort standardized expression and mean signature scoring. The primary recurrent-miscarriage decidual inflammatory macrophage/APC signature showed complete 30-gene coverage in all three external cohorts and positive case-associated shifts in each dataset. Cohort-level deltas were +1.019, +0.282 and +0.941, giving a mean delta of +0.747 and positive directional recurrence in 3/3 cohorts. Secondary decidua immune-stromal, villi trophoblast/epithelial and recurrent-miscarriage villi structural/trophoblast signatures showed weaker, modest or context-dependent behaviour. These findings support exploratory cross-cohort signature recurrence of an inflammatory decidual macrophage/APC tissue-state axis in recurrent pregnancy loss. The result is observational and hypothesis-generating; it does not establish causality, diagnostic performance, clinical utility or standalone biomarker validation.

Introduction

Infertility and pregnancy loss impose substantial clinical, psychological and societal burdens. The World Health Organization estimates infertility as a common global reproductive-health condition, while recurrent pregnancy loss (RPL) represents a distinct clinical problem requiring its own evaluation and management pathway [1-3]. Contemporary guidelines differ in definitions and thresholds, but both ESHRE and ASRM emphasize the heterogeneity of RPL and the need to evaluate established contributors such as genetic abnormalities, uterine factors, endocrine disease, antiphospholipid syndrome and other maternal or embryonic causes [2,3]. Beyond these recognized categories, miscarriage and recurrent miscarriage also carry physical, psychological and future obstetric consequences, making mechanistic understanding of early pregnancy failure a human-health priority rather than a narrow fertility issue [4,5].

41 The early maternal-fetal interface provides a biologically plausible site for unexplained or incompletely
 42 explained pregnancy loss. Human decidualization transforms endometrial stromal fibroblasts into specialized
 43 decidual cells, coordinates glandular and vascular remodelling, and creates a tissue matrix in which immune
 44 competence must coexist with tolerance of fetal trophoblast [6,7]. Single-cell atlases of first-trimester decidua
 45 and placenta have shown that this interface is a structured multicellular ecosystem containing decidual stromal
 46 cells, macrophages, natural killer cells, T cells, endothelial and mural cells, epithelial compartments and
 47 invading trophoblasts [8,9]. Trophoblast-decidual immune crosstalk, local trophoblast recognition by uterine
 48 immune cells and regulatory T-cell signatures are therefore central to early placentation and immune balance
 49 [10-13].

50 RPL has repeatedly been linked to immune dysregulation, but the direction, cellular basis and translational
 51 meaning of immune signals remain unresolved. Single-cell and immune-profiling studies have reported altered
 52 distributions of decidual leukocyte subsets in RPL, unexplained RPL and recurrent miscarriage, including
 53 changes in NK-cell, T-cell and myeloid compartments [14-17]. Decidual macrophages are particularly relevant
 54 because they participate in phagocytosis, tissue remodelling, angiogenesis, trophoblast interaction, cytokine
 55 production and immune regulation [18,22,23]. Disease-focused studies have associated RPL or recurrent
 56 spontaneous abortion with altered decidual macrophage states, CCR1+ macrophage dysfunction, M1-like
 57 polarization and macrophage-related diagnostic signatures [18-21,30]. Complement-associated macrophage
 58 biology is also plausible at the maternal-fetal interface because C1q and complement-related pathways
 59 contribute to implantation-site inflammation, trophoblast invasion and tissue remodelling, although excessive or
 60 mislocalized complement activity can be harmful [24-27].

61 Despite this biological rationale, public transcriptomic evidence remains hard to integrate. Bulk decidual RNA
 62 profiles can detect disease-associated expression differences, but they cannot directly distinguish increased
 63 macrophage abundance from macrophage activation, altered decidual stromal composition, trophoblast
 64 admixture, tissue injury or platform-specific artefact. Many public RPL studies are small, and single-cohort
 65 differential-expression lists may not recur across datasets [31-36]. Single-cell-derived signature projection offers
 66 an intermediate strategy: derive interpretable signatures from a cellular reference, then test whether those
 67 signatures recur directionally across independent bulk cohorts. This approach is different from diagnostic
 68 classifier construction; it evaluates whether a biologically coherent tissue-state axis is detectable beyond the
 69 discovery dataset.

70 The present study addresses this gap by testing the cross-cohort recurrence of GSE214607-derived recurrent-
 71 miscarriage and tissue-context signatures across three independent bulk decidua cohorts. The aims were: first, to
 72 derive compact and interpretable projection-ready signatures from a disease single-cell/pseudobulk reference;
 73 second, to determine whether these signatures show adequate gene coverage when projected into independent
 74 bulk decidua datasets; and third, to assess whether a recurrent-miscarriage decidual inflammatory
 75 macrophage/APC signature shows concordant case-associated recurrence across cohorts. Figure 1 summarizes
 76 the workflow, Table 1 summarizes the datasets, and Supplementary Table S1 documents metadata inclusion.

77 **Materials and Methods**

78 This was a retrospective computational reanalysis of public de-identified transcriptomic datasets. No new
 79 participants were recruited, and no new identifiable human data were generated. The analysis was designed
 80 around cross-cohort signature recurrence rather than biomarker discovery or diagnostic classification. The
 81 overall workflow is summarized in Fig. 1.

82 GSE214607 was used as the primary disease single-cell reference because it contains first-trimester decidua and
 83 villi samples from control and recurrent-miscarriage groups and was originally generated to explore

macrophage-associated recurrent miscarriage signals [28-30]. Raw 10x Genomics matrix files were imported as sparse count matrices, checked for matrix/barcode/feature consistency and assigned to control decidua, recurrent-miscarriage decidua, control villi or recurrent-miscarriage villi categories. Per-cell quality-control metrics included total counts and detected genes. Cells passing predefined feature thresholds were retained for sample-level aggregation.

To avoid pseudoreplication, retained GSE214607 cells were aggregated into sample-level pseudobulk profiles by summing counts within each sample. Library-size-normalized log2 counts per million were then calculated. Pseudobulk aggregation was used for signature derivation because cells from the same individual are not independent biological replicates and single-cell differential-expression procedures can underestimate variance if sample structure is ignored [37,38]. The pseudobulk profiles were used only to derive exploratory signatures and not to make formal differential-expression claims.

Candidate signatures were derived from three contrast families: recurrent-miscarriage decidua versus control decidua, recurrent-miscarriage villi versus control villi and decidua versus villi. Candidate genes were ranked by direction and magnitude of expression shift. Lists were then filtered for interpretability: low-information noncoding, housekeeping, ribosomal, mitochondrial and technical features were excluded unless they had explicit maternal-fetal-interface relevance. The final projection-ready framework retained four 30-gene signatures: a recurrent-miscarriage decidual inflammatory macrophage/APC signature, a decidua immune-stromal identity signature, a villi trophoblast/epithelial identity signature and a recurrent-miscarriage villi structural/trophoblast signature (Table 2; Supplementary Table S2).

Three external bulk decidua datasets were used for projection: GSE161969, GSE113790 and GSE178535 [31-36]. GSE161969 was processed from a public gene-symbol FPKM matrix and transformed as $\log_2(\text{FPKM} + 1)$. GSE113790 was imported as a processed FPKM expression matrix and curated at the gene-symbol level. GSE178535 was analysed from its public normalized log2 expression matrix. Sample labels were manually curated to harmonize control and RPL/RSA case categories while preserving dataset-specific terminology (Supplementary Table S1).

For each cohort and signature, genes were intersected with the cohort expression matrix, and coverage was recorded before scoring. Within each cohort, each gene was standardized across samples. Signature scores were then calculated as the mean standardized expression of the matched signature genes. This simple mean-z scoring was chosen to preserve interpretability and avoid cross-platform absolute-expression assumptions; related single-sample gene-set approaches use analogous logic to summarize coordinated expression at the sample level [39].

Cross-cohort recurrence was evaluated using technical coverage, case-minus-control signature-score deltas and directional concordance. Complete or near-complete signature coverage was required for confident interpretation. For each signature and dataset, mean scores were calculated for controls and cases, and case-minus-control deltas were computed. Directional recurrence was defined as a consistent sign of case-associated shift across cohorts. Wilcoxon rank-sum tests and Benjamini-Hochberg-adjusted values were calculated descriptively but were not used as the primary criterion because the external cohorts were small [40]. Full projection-score and metadata files are listed in Supplementary Tables S2-S4.

Results

The analysis integrated one disease single-cell/pseudobulk reference and three independent bulk decidua cohorts. GSE214607 provided the signature-derivation reference, while GSE161969, GSE113790 and GSE178535 provided independent projection settings (Table 1; Fig. 1). The use of three external cohorts

126 allowed the study to prioritize directional recurrence across independent datasets rather than within-dataset
127 statistical significance.

128 Pseudobulk contrasts from GSE214607 generated both recurrent-miscarriage-associated and tissue-context
129 signatures. After interpretability filtering, four compact signatures were retained (Table 2). The recurrent-
130 miscarriage decidual inflammatory macrophage/APC signature was designated the primary signature because it
131 was derived from the disease-associated decidual contrast and included coherent myeloid, antigen-presentation
132 and complement-associated genes. The signature included HLA-DRA, HLA-DPA1, HLA-DPB1, CD74,
133 S100A8, S100A9, AIF1, MS4A7, C1QA, C1QB, C1QC, CTSZ, SPP1 and GRN, supporting interpretation as a
134 macrophage/APC-rich inflammatory tissue-state signature rather than an arbitrary gene list (Table 2;
135 Supplementary Table S2; Fig. 2).

136 The primary macrophage/APC signature showed complete technical coverage in all external bulk cohorts. All
137 30 genes were detected in GSE161969, GSE113790 and GSE178535, supporting reliable projection across
138 different processed expression matrices. Secondary signatures also had high coverage, but the macrophage/APC
139 signature was the most technically robust primary disease-associated signature (Supplementary Table S3;
140 Supplementary Fig. 1).

141 The macrophage/APC signature showed a positive case-associated shift in all three external cohorts. The case-
142 minus-control delta was +1.019 in GSE161969, +0.282 in GSE113790 and +0.941 in GSE178535. This yielded
143 positive directional recurrence in 3/3 cohorts, a mean delta of +0.747 and a median delta of +0.941 (Table 3;
144 Fig. 3). Because the cohorts were small, this result is interpreted as exploratory directional recurrence rather
145 than a validated diagnostic effect. Nevertheless, the consistency of direction across three datasets supports cross-
146 cohort signature recurrence of the decidual inflammatory macrophage/APC axis.

147 Secondary signatures showed weaker or context-dependent behaviour. The decidua immune-stromal identity
148 signature was positive in all three cohorts but with smaller deltas (+0.573, +0.206 and +0.031). The villi
149 trophoblast/epithelial identity signature showed modest positive shifts (+0.134, +0.406 and +0.156), while the
150 recurrent-miscarriage villi structural/trophoblast signature was mixed (-0.097, +0.245 and -0.007). Thus, not all
151 GSE214607-derived signatures showed the same behaviour in decidua-focused external cohorts (Table 3; Fig.
152 4). This pattern supports tissue-context-aware interpretation and argues against a nonspecific projection artefact
153 shared equally by all signatures.

154 Together, the results identify a single-cell-derived recurrent-miscarriage decidual inflammatory
155 macrophage/APC signature with complete 30-gene coverage and positive directional recurrence across three
156 independent bulk decidua cohorts. The main result is a reproducible tissue-state signal, not a causal proof,
157 clinical diagnostic model or biomarker validation. Intermediate single-cohort and two-cohort analyses are
158 retained for reproducibility in Supplementary Fig. 2.

159 Discussion

160 This study provides exploratory evidence that a GSE214607-derived recurrent-miscarriage decidual
161 inflammatory macrophage/APC signature recurs directionally across three independent bulk decidua cohorts.
162 The study was designed to answer a conservative reproducibility question: does an interpretable single-cell-
163 derived tissue-state signature recur across datasets, tissues and preprocessing contexts? The answer was
164 strongest for the macrophage/APC signature, which showed complete gene coverage and positive case-
165 associated shifts in GSE161969, GSE113790 and GSE178535. This directly addresses the first and second aims
166 by demonstrating that compact signature derivation and technically robust projection are feasible, and addresses
167 the third aim by showing concordant cross-cohort recurrence of the primary inflammatory myeloid signature.

Scientifically, the result fits a growing literature in which RPL is viewed not only as an embryonic or anatomical problem but also as a disorder that may involve disturbed decidual immune balance. Guidelines emphasize that RPL is clinically heterogeneous and that many cases remain unexplained after standard evaluation [2,3]. The Lancet miscarriage series further frames recurrent miscarriage as a high-burden condition that requires better biological understanding and improved care pathways [4,5]. The present data do not redefine RPL clinically, but they add a reproducible transcriptomic tissue-state observation that may help organize future mechanistic work.

The macrophage/APC signature is biologically plausible because macrophages are positioned at the interface between immune tolerance, tissue repair, trophoblast interaction, angiogenesis and inflammation. Single-cell studies of normal early pregnancy show that decidua contains diverse immune and stromal populations whose interactions are essential for successful placentation [8-11]. Disease-focused single-cell and immune-profiling studies have independently reported altered decidual leukocyte states in RPL or unexplained RPL, including myeloid, NK-cell and T-cell alterations [14-17]. The current finding is therefore similar in direction to published immune-landscape studies: it supports the presence of an altered decidual immune tissue state. Its specific contribution is methodological and cross-dataset: it shows that a compact macrophage/APC signature can be projected and recovered across three independent bulk cohorts rather than inferred from one dataset alone.

The signature also agrees with macrophage-focused RPL literature, but it should be interpreted carefully. Reviews and primary studies have emphasized that decidual macrophages cannot be reduced to a simple M1/M2 dichotomy and may adopt hybrid, context-dependent states [18,22,23]. CCR1+ macrophage dysfunction, M1-like macrophage polarization and impaired anti-inflammatory macrophage programming have been reported in RPL or recurrent spontaneous abortion contexts [19-21]. The present signature includes antigen-presentation genes, inflammatory myeloid markers and C1q-associated macrophage genes, so it is coherent with these reports. However, the bulk projection design cannot determine whether the signal reflects more macrophages, a different macrophage state, increased antigen presentation, complement-associated tissue remodelling or secondary response to tissue injury. This is a key difference from functional macrophage studies: our analysis supports recurrence of a signature, not the mechanism that generates it.

Complement-related genes in the primary signature, especially C1QA, C1QB and C1QC, add biological interest. Complement pathways are active at the implantation site and can contribute to inflammation, trophoblast invasion, vascular remodelling and clearance of apoptotic material [24-26]. Recent decidual bioinformatics work has also linked complement/coagulation modules and macrophage abundance to RPL datasets [27]. The similarity is useful because it places the current C1q-containing macrophage/APC signature within a broader complement-inflammation literature. The difference is equally important: the current study does not claim that complement activation causes RPL or that C1q is independently pathogenic. C1q genes may mark macrophage identity, trophoblast/stromal interaction, tissue-remodelling macrophages or local innate immune activation, and these possibilities require spatial and protein-level validation.

Methodologically, the study contributes a cross-cohort recurrence-oriented way to reuse public reproductive transcriptomic data. Single-cell datasets provide cellular resolution but are often small, whereas bulk datasets provide independent cohorts but have mixed-cell composition. Pseudobulk summarization respects biological replication in single-cell analysis and reduces false confidence caused by treating cells as independent samples [37,38]. Mean standardized signature scoring is transparent and interpretable, and the use of directional concordance across three cohorts is appropriate for small public datasets where individual P values are unstable [39,40]. This differs from diagnostic classifier studies, including the original macrophage-marker model associated with GSE214607 [30]. The present analysis deliberately avoids claiming a predictive model and instead asks whether the biological signature itself recurs across cohorts.

211 The weaker behaviour of the secondary signatures is informative. If every GSE214607-derived signature shifted
212 similarly across all external datasets, the result could reflect a nonspecific batch or disease-projection artefact.
213 Instead, the decidua immune-stromal signature was positive but weaker, villi/trophoblast-related signatures were
214 modest or mixed, and the macrophage/APC signature was the most consistent. This supports the interpretation
215 that decidua-relevant inflammatory myeloid biology, rather than generic signature projection, drives the primary
216 result. It also reinforces the importance of tissue context: a villi-derived signature should not necessarily recur
217 strongly into decidua-enriched bulk cohorts.

218 Translationally, the finding should be treated as a prioritization result. It nominates a decidual inflammatory
219 macrophage/APC axis for future validation in larger, better annotated cohorts. Useful follow-up studies would
220 include euploid versus aneuploid loss stratification, gestational-age matching, standardized tissue sampling,
221 histopathology, fetal karyotype, endocrine and immune metadata, and prospective replication. Spatial
222 transcriptomics, multiplex immunofluorescence, imaging mass cytometry or targeted immunohistochemistry
223 could test whether the signature localizes to specific decidual niches and whether it reflects macrophage
224 abundance, activation state, antigen-presentation activity, complement-associated remodelling or interaction
225 with trophoblast and stromal compartments.

226 Several limitations remain. First, the external datasets are small, so the evidence rests on directionality across
227 cohorts rather than cohort-level statistical validation. Second, public metadata are incomplete, and clinical labels
228 such as RPL, RM, RSA and unexplained RSA may not represent identical biological entities. Third, bulk
229 signature scores cannot distinguish cell abundance from cell state. Fourth, GSE214607 itself is small and
230 disease-enriched, so derived signatures may capture disease-context signatures rather than universal macrophage
231 identities. Fifth, the reference list and biological framing in v2 are broader than v1.1, but the central empirical
232 result is unchanged: a 30-gene macrophage/APC signature shows positive recurrence across three cohorts.

233 In conclusion, this study identifies a single-cell-derived recurrent-miscarriage decidual inflammatory
234 macrophage/APC signature that recurs directionally across three independent bulk decidua datasets. The revised
235 evidence framework positions the result within RPL clinical burden, maternal-fetal-interface biology, decidual
236 macrophage literature, complement-associated inflammation and public-data cross-cohort recurrence
237 methodology. The finding is biologically coherent and methodologically useful, but it remains exploratory and
238 should not be interpreted as causal, diagnostic or clinically actionable without independent spatial, protein-level
239 and prospective validation.

240 Conclusion

241 This study provides exploratory evidence that a single-cell-derived recurrent-miscarriage decidual inflammatory
242 macrophage/APC signature recurs directionally across three independent bulk decidua transcriptomic cohorts.
243 The finding supports the use of single-cell-derived cross-cohort signature analysis to identify reproducible
244 tissue-state signatures in recurrent pregnancy loss. The macrophage/APC axis is biologically coherent and
245 reproducible at the cross-cohort signature level, but it should not yet be considered causal, diagnostic or
246 clinically actionable.

247 Data Availability Statement

248 Figure-caption support files, reproducibility-audit outputs and release documentation supporting manuscript
249 assembly are summarized in Supplementary Table S4.

250 All datasets analysed in this study are publicly available from GEO or associated public sequence repositories
251 under accession numbers GSE214607, SRP400574, GSE161969, SRP293708, GSE113790, SRP143517 and

252 GSE178535. Derived signatures, curated metadata summaries, projection scores, cross-cohort recurrence
253 summaries, figures, manuscript drafts, QC reports and reproducibility documentation are available at
254 <https://github.com/rmaghembe1/rpl-decidua-transportability>. Large raw downloads and regenerable intermediate
255 objects are not redistributed because the source datasets remain publicly available.

256 **Author contributions**

257 Reuben S. Maghembe: conceptualisation, methodology, software, formal analysis, investigation, data curation,
258 visualisation, writing - original draft, writing - review and editing, project administration.

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265 GSE113790 and GSE178535 datasets. Public data sharing and open-source scientific software made this
266 integrative transcriptomic reanalysis possible.

267 **Competing interests**

268 The author declares no competing interests.

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362 **Supporting information captions**

363 Supplementary Table S1. Dataset metadata and sample inclusion framework. Curated sample metadata, tissue
364 source, disease group, analysis role and inclusion/exclusion status for GSE214607, GSE161969, GSE113790
365 and GSE178535.

366 Supplementary Table S2. GSE214607-derived projection-ready signature gene sets. Full gene lists for the four
367 projection-ready signatures.

368 Supplementary Table S3. Three-cohort signature projection and cross-cohort recurrence statistics. Signature
369 scores, gene-overlap metrics, cohort-specific case-minus-control deltas and cross-cohort concordance
370 summaries.

371 Supplementary Table S4. Figure-caption support and reproducibility audit. Figure-source tables, script/output
372 inventory, release-manifest notes and interpretation-boundary documentation.

373 Supplementary Figure S1. Dataset processing, quality-control and signature-coverage summary.

374 Supplementary Figure S2. Intermediate single-cohort and two-cohort projection summaries supporting the final
375 three-cohort analysis.

376 **Figure legends**

377 Figure 1. Study design and analytical workflow

378 Single-cell transcriptomic cross-cohort recurrence workflow for recurrent pregnancy loss decidua analysis.
379 GSE214607 was used as a disease single-cell/pseudobulk reference and GSE161969, GSE113790 and
380 GSE178535 were used as independent bulk decidua projection cohorts. Candidate signatures were derived from
381 pseudobulk contrasts, filtered for interpretability and projected into each bulk cohort using within-cohort
382 standardized expression. Cross-cohort recurrence was evaluated by coverage, case-minus-control deltas and
383 directional concordance.

384 Figure 2. GSE214607-derived projection-ready tissue-state signatures

385 Derivation of projection-ready recurrent-miscarriage and tissue-context signatures from GSE214607. Four
386 compact signatures were retained: recurrent-miscarriage decidual inflammatory macrophage/APC up, decidua
387 immune-stromal identity, villi trophoblast/epithelial identity and recurrent-miscarriage villi
388 structural/trophoblast.

389 Figure 3. Three-cohort recurrence of the macrophage/APC signature

390 The recurrent-miscarriage decidual inflammatory macrophage/APC signature showed complete 30/30 gene
391 coverage and positive case-associated shifts in GSE161969, GSE113790 and GSE178535. Deltas were +1.019,
392 +0.282 and +0.941, respectively.

393 Figure 4. Cross-cohort delta summary for all projected signatures

394 Case-minus-control signature-score deltas for four GSE214607-derived signatures across external cohorts. The
395 macrophage/APC signature showed the strongest and most consistent positive directional recurrence;
396 villi/trophoblast-associated signatures were weaker or mixed.

397 **Tables**

398 Table 1. Public datasets used in the analysis

Dataset	Assay/data type	Tissue/source	Groups used	Role in study	Key processing	Main limitation
GSE214607	Single-cell RNA-seq / pseudobulk reference	First-trimester decidua and villi	Control and recurrent miscarriage	Disease single-cell reference; signature derivation	Raw 10x matrices imported, QC-filtered and aggregated to pseudobulk profiles	Limited sample numbers; exploratory signature derivation
GSE161969	Bulk RNA-seq processed expression	Decidua	Control and RPL	External projection cohort	Gene-symbol FPKM matrix; log2(FPKM + 1); within-cohort scoring	Very small cohort; limited metadata
GSE113790	Bulk transcriptomic processed expression	Decidua	Control and RSA	External projection cohort	FPKM matrix imported; labels curated; within-cohort scoring	Very small cohort; metadata harmonization constraints
GSE178535	Bulk transcriptomic processed expression	Decidua	Control and RSA	External projection cohort	Public normalized log2 matrix; labels curated; within-cohort scoring	Very small cohort; platform/preprocessing differences

400 Table 2. GSE214607-derived projection-ready signatures

Code	Short label	Source contrast	Genes	Biological interpretation	Use
M/APC-RM	RM decidua inflammatory macrophage/APC up	RM decidua vs control decidua	30	Antigen-presentation, inflammatory myeloid and complement-associated macrophage/APC signature	Primary signature
DEC-IS	Decidua immune-stromal identity	Decidua vs villi	30	Decidua-enriched immune and stromal tissue-context signature	Secondary comparator
VIL-TE	Villi trophoblast/epithelial identity	Villi vs decidua	30	Villi/trophoblast/epithelial tissue-context signature	Tissue-context comparator
VIL-ST-RM	RM villi structural/trophoblast signature	RM villi vs control villi	30	Villi structural/trophoblast-associated signature	Secondary tissue-context comparator

401

402 Table 3. Three-cohort signature recurrence summary

Signature	GSE161969 delta	GSE113790 delta	GSE178535 delta	Positive cohorts	Minimum coverage	Concordance interpretation
RM decidua macrophage/APC	+1.019	+0.282	+0.941	3/3	100%	Concordant positive directional signal
Decidua immune- stromal	+0.573	+0.206	+0.031	3/3	96.7%	Positive but weaker/context-dependent
Villi trophoblast/ epithelial	+0.134	+0.406	+0.156	3/3	100%	Modest positive; tissue- context dependent
RM villi structural/ trophoblast	-0.097	+0.245	-0.007	1/3	96.7%	Mixed/weak; not recurrent in decidua cohorts

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